MSDM5004 Spring 2026
Homework 2 Part I
Due Mar. 15

1. Find the first two iterations using (1) the Jacobi method and (2) the Gauss-Seidel method for the following linear system, starting from $\mathbf{x}^{(0)} = \mathbf{0}$.

$$\begin{aligned} 8x_1 - 3x_2 + 2x_3 =& 20, \\ 4x_1 + 11x_2 - x_3 =& 33, \\ 6x_1 + 3x_2 + 12x_3 =& 36. \end{aligned}$$

2. Write codes using MATLAB (or other programming language) to solve the following linear system using (1) the Jacobi method and (2) the SOR method with $\omega = 1.3$. The initial estimate is $\mathbf{x}^{(0)} = (0, 0, 0, 0)^T$. Stop the iterations until the l_∞ norm $\|\mathbf{x}^{(k)} - \mathbf{x}^{(k-1)}\|_\infty \leq 10^{-5}$.

$$\begin{aligned} 4x_1 + x_2 + x_3 + x_4 =& 1, \\ x_1 + 4x_2 + x_3 + x_4 =& 1, \\ x_1 + x_2 + 4x_3 + x_4 =& 1, \\ x_1 + x_2 + x_3 + 4x_4 =& 1. \end{aligned}$$